

Lab 1

Introduction

```
data('arbuthnot', package='openintro')
```

```
arbuthnot
```

```
# A tibble: 82 x 3
  year  boys girls
  <int> <int> <int>
1  1629  5218  4683
2  1630  4858  4457
3  1631  4422  4102
4  1632  4994  4590
5  1633  5158  4839
6  1634  5035  4820
7  1635  5106  4928
8  1636  4917  4605
9  1637  4703  4457
10 1638  5359  4952
# i 72 more rows
```

```
glimpse(arbuthnot)
```

```
Rows: 82
```

```
Columns: 3
```

```
$ year  <int> 1629, 1630, 1631, 1632, 1633, 1634, 1635, 1636, 1637, 1638, 1639~
```

```
$ boys  <int> 5218, 4858, 4422, 4994, 5158, 5035, 5106, 4917, 4703, 5359, 5366~
```

```
$ girls <int> 4683, 4457, 4102, 4590, 4839, 4820, 4928, 4605, 4457, 4952, 4784~
```

Exercise 1

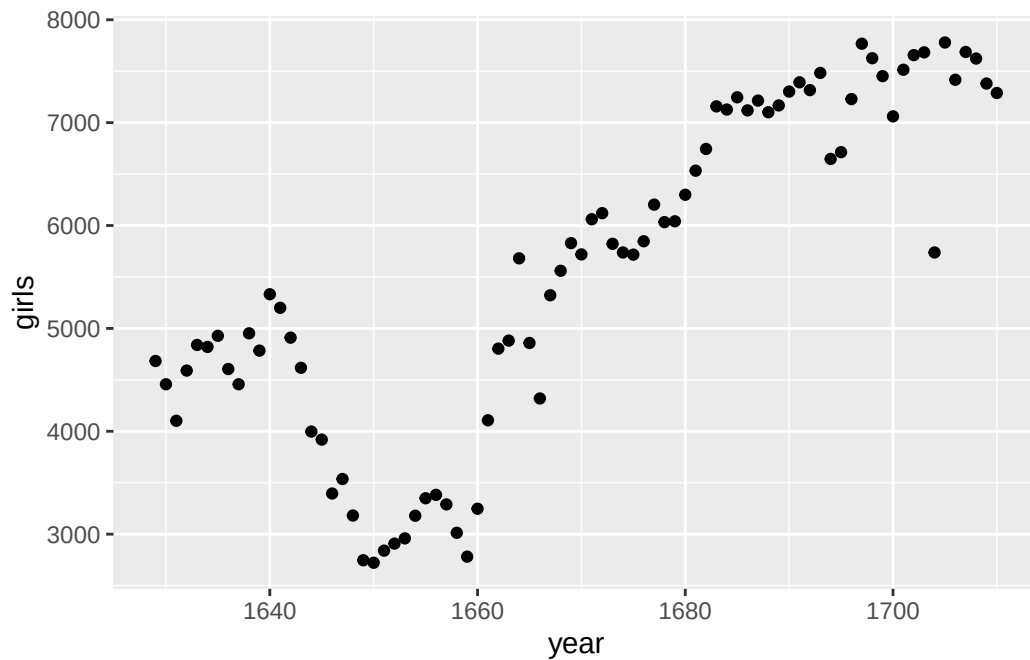
The command to view the counts of the girls baptized will be `sum(arbuthnot$girls)` output seen below

```
sum(arbuthnot$girls)
```

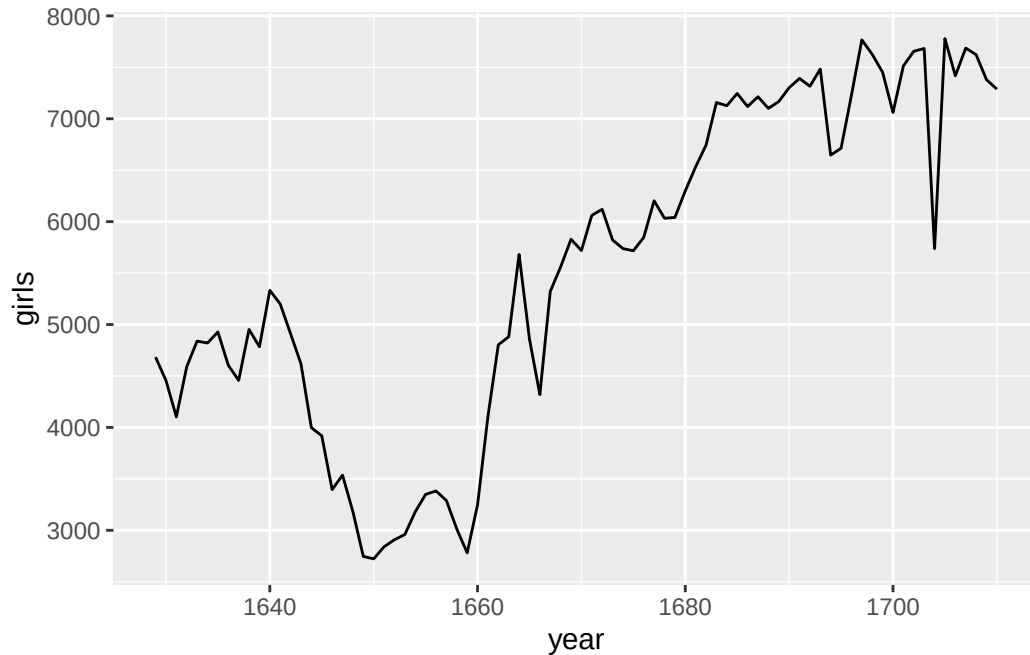
```
[1] 453841
```

Data visualization

```
ggplot(data = arbuthnot, aes(x = year, y = girls)) +  
  geom_point()
```



```
ggplot(data = arbuthnot, aes(x = year, y = girls)) +  
  geom_line()
```



Exercise 2

Analyzing the previous plots shows us that the amount of baptisms for girls shrunk dramatically between 1640 and 1650. However, in 1660 the amount of girls baptized began to shoot up and had a fairly consistent trend into the early 1700s with a few down years that never met the lows of pre 1660.

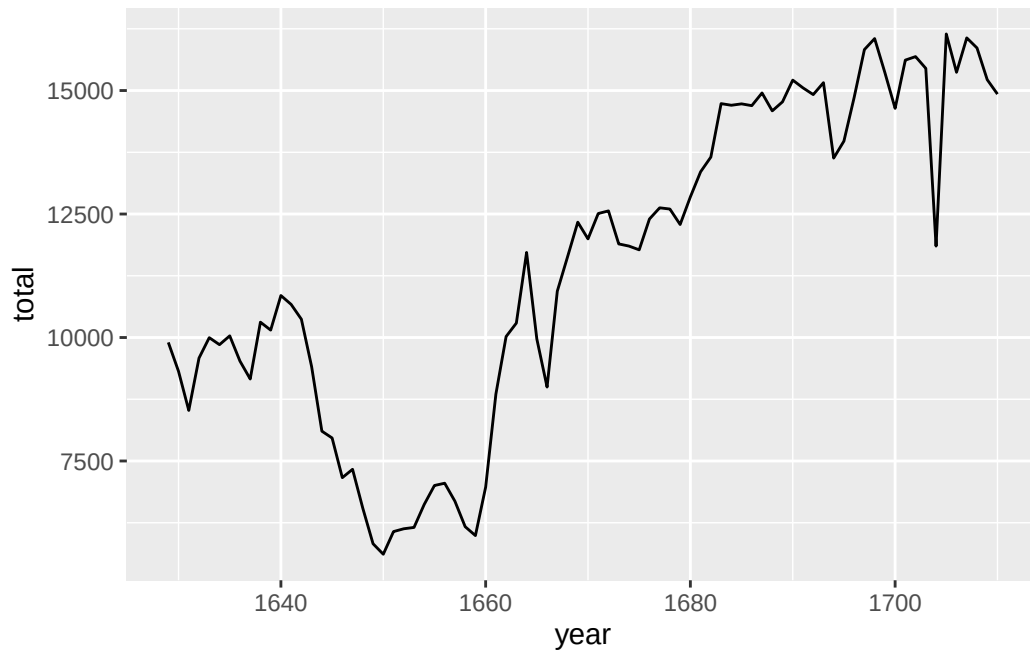
R as a calculator

```
arbuthnot$boys + arbuthnot$girls
```

```
[1] 9901 9315 8524 9584 9997 9855 10034 9522 9160 10311 10150 10850
[13] 10670 10370 9410 8104 7966 7163 7332 6544 5825 5612 6071 6128
[25] 6155 6620 7004 7050 6685 6170 5990 6971 8855 10019 10292 11722
[37] 9972 8997 10938 11633 12335 11997 12510 12563 11895 11851 11775 12399
[49] 12626 12601 12288 12847 13355 13653 14735 14702 14730 14694 14951 14588
[61] 14771 15211 15054 14918 15159 13632 13976 14861 15829 16052 15363 14639
[73] 15616 15687 15448 11851 16145 15369 16066 15862 15220 14928
```

```
# |> as a piping tool is just a bit easier to type
arbuthnot <- arbuthnot |>
  mutate(total = boys + girls)

ggplot(data = arbuthnot, aes(x = year, y = total)) +
  geom_line()
```



```
5218 / 4683
```

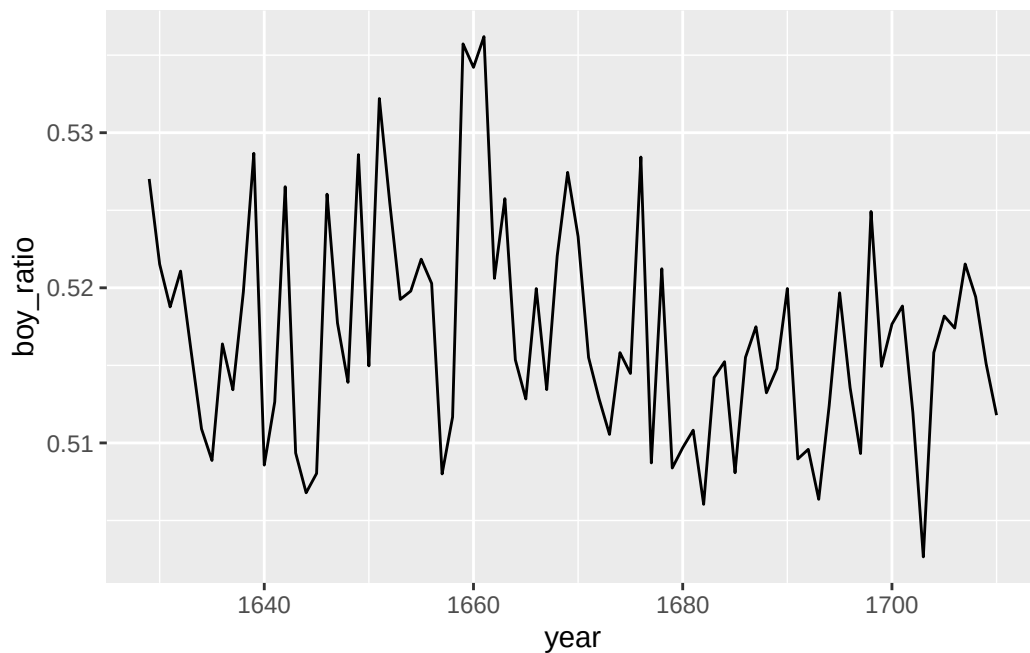
```
[1] 1.114243
```

```
5218 / (5218 + 4683)
```

```
[1] 0.5270175
```

```
arbuthnot <- arbuthnot |>
  mutate(boy_ratio = boys / total)

ggplot( data = arbuthnot, aes(x = year, y = boy_ratio)) +
  geom_line()
```



```
mean(arbuthnot$boy_ratio)
```

```
[1] 0.5169751
```

```
max(arbuthnot$boy_ratio)
```

```
[1] 0.5361942
```

Exercise 3

When looking at the ratio of boys to girls I can see that it is consistently above the 50 percent mark with the mean being 51.6% and reaching a peak of 53.6%

More Practice

```
data('present', package='openintro')  
  
arbuthnot |>  
  summarize(min=min(boys), max=max(boys))
```

```
# A tibble: 1 x 2
  min    max
<int> <int>
1  2890  8426
```

```
present |>
  summarize(min=min(year), max=max(year))
```

```
# A tibble: 1 x 2
  min    max
<dbl> <dbl>
1  1940  2002
```

```
years<-present |>
  summarize(min=min(year), max=max(year))

dim(present)
```

```
[1] 63  3
```

```
names(present)
```

```
[1] "year" "boys" "girls"
```

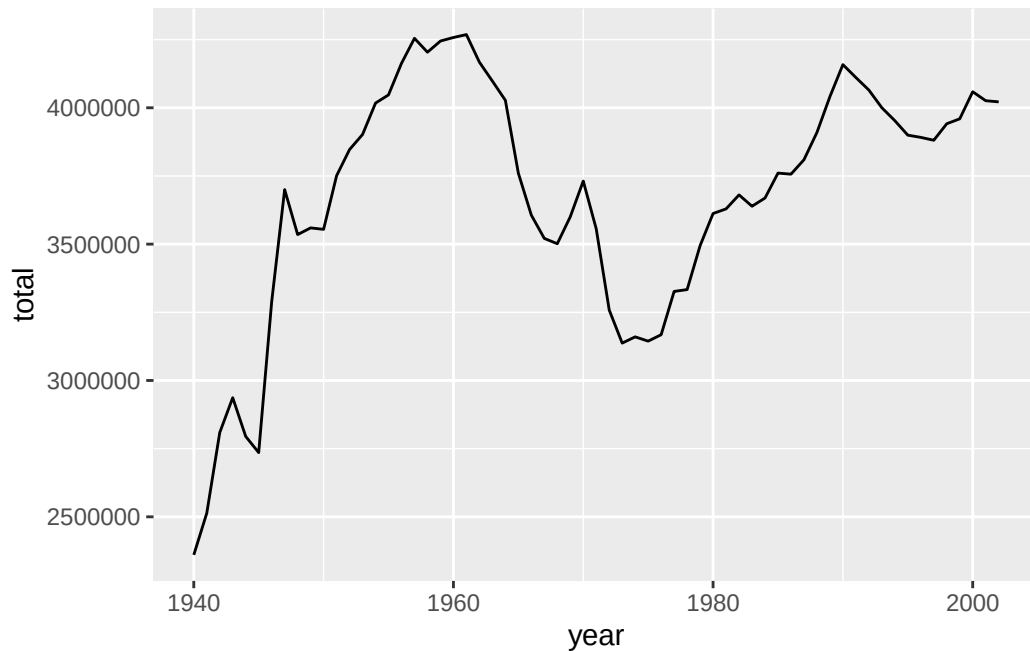
Exercise 4

The years included in the dataset are 1940, 2002, the dimensions of the dataset are 63, 3, and the names of the columns in the dataset are year, boys, girls

Exercise 5

```
present$total <- present$boys + present$girls

ggplot(data = present, aes(x = year, y = total) ) +
  geom_line()
```

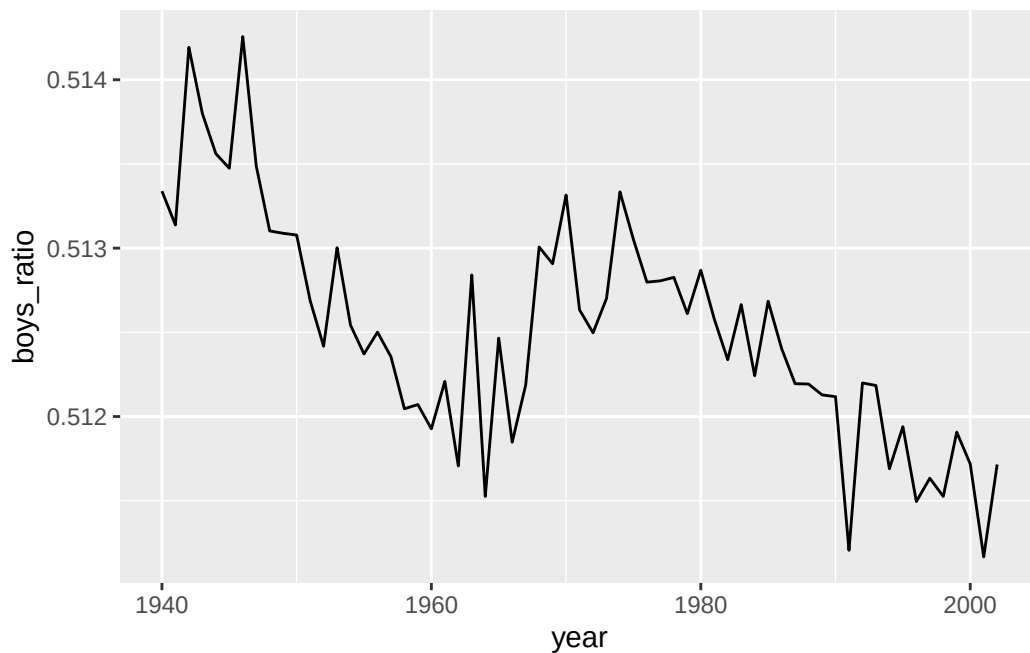


The magnitude is on a completely different scale when comparing the 1600s dataset to the modern day dataset. The modern day dataset is several times larger than the Arbutnot dataset

Exercise 6

```
present <- present |>
  mutate( boys_ratio = boys / total )

ggplot( data = present, aes( x = year, y = boys_ratio)) +
  geom_line()
```



The margin in which boys are born more often than girls is much lower, but it seems to show a trend that boys have continued to be born slightly more than girls.

Exercise 7

```
present[ which(present$total == max(present$total)), 'year']
```

```
# A tibble: 1 x 1
  year
  <dbl>
1 1961
```

#or to follow the exercise suggestion

```
present |>
  arrange(desc(total))
```

```
# A tibble: 63 x 5
  year    boys girls total boys_ratio
  <dbl> <dbl> <dbl> <dbl>    <dbl>
```

1	1961	2186274	2082052	4268326	0.512
2	1960	2179708	2078142	4257850	0.512
3	1957	2179960	2074824	4254784	0.512
4	1959	2173638	2071158	4244796	0.512
5	1958	2152546	2051266	4203812	0.512
6	1962	2132466	2034896	4167362	0.512
7	1956	2133588	2029502	4163090	0.513
8	1990	2129495	2028717	4158212	0.512
9	1991	2101518	2009389	4110907	0.511
10	1963	2101632	1996388	4098020	0.513

i 53 more rows

The highest number of births come from the baby boomer generation of 1961